

KAKTOVIK, AK, USA

WMO: 700860

Lat: 70.134N

Lon: 143.577W

Elev: 2

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27401

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.8	-32.1	-43.7	0.4	-34.8	-39.3	0.6	-31.2	48.5	-2.5	44.2	-0.4	5.8	260	1.083

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	59.9	55.2	54.8	51.7	51.1	48.8	55.7	59.2	51.9	54.6	48.9	50.9	9.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.6	61.2	58.6	50.0	53.4	53.7	47.4	48.3	50.4	23.6	59.5	21.4	54.5	19.6	50.8	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 40.6	34.6	29.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-34.8	61.7	5.6	10.2	-38.8	69.0	-42.1	75.0	-45.2	80.7	-49.3	88.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	-8.2	-10.0	-8.0	7.8	25.4	35.1	43.7	42.2	35.3	23.5	6.3	-2.4	
	DBStd	22.29	13.25	13.52	14.78	13.29	8.08	5.06	5.41	7.13	4.89	7.79	11.07	12.87	
	HDD50	12436	1806	1681	1798	1269	762	449	208	262	441	823	1311	1626	
	HDD65	17876	2271	2101	2263	1715	1227	898	661	710	891	1288	1761	2090	
	CDD50	40	0	0	0	5	0	0	12	22	0	0	0	1	
	CDD65	6	0	0	0	1	0	0	0	5	0	0	0	0	
	CDH74	81	0	0	0	9	0	0	0	64	0	0	0	8	
	CDH80	25	0	0	0	3	0	0	0	18	0	0	0	4	
Wind	WSAvg	12.5	13.5	12.7	11.5	11.3	11.5	11.4	10.8	11.5	12.7	15.1	14.6	13.9	
Precipitation	PrecAvg	4.5	0.2	0.1	0.2	0.1	0.2	0.4	0.8	0.9	0.6	0.6	0.3	0.2	
	PrecMax	9.0	1.8	0.7	1.5	0.4	1.2	1.3	3.0	2.2	1.7	1.4	1.1	0.7	
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	2.1	0.4	0.2	0.3	0.1	0.3	0.4	0.8	0.6	0.4	0.4	0.3	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	30.2	45.9	69.6	40.9	55.3	64.0	82.5	53.1	43.4	33.1	47.5	
		MCWB	35.4	29.8	42.3	63.9	38.0	50.0	56.7	80.4	47.7	40.1	31.4	45.5	
	2%	DB	22.7	21.6	31.7	37.9	37.8	48.5	58.7	59.0	48.1	36.7	27.6	27.5	
		MCWB	21.9	20.7	30.0	34.1	35.9	45.5	54.4	54.9	44.7	35.3	26.4	25.7	
	5%	DB	13.8	14.2	20.0	29.2	36.1	44.5	54.5	54.3	44.4	34.0	24.7	19.0	
		MCWB	13.0	13.3	19.2	27.7	34.5	42.1	51.9	51.5	42.2	33.0	23.5	18.0	
	10%	DB	9.2	8.4	12.1	24.7	34.5	40.9	51.2	50.9	41.5	32.2	21.4	14.6	
		MCWB	8.3	7.3	11.0	23.3	33.2	39.3	49.2	48.8	40.2	31.4	20.7	13.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	29.8	42.8	63.5	38.4	50.0	57.9	80.5	48.4	40.5	31.5	44.6
			MCDB	39.2	30.8	45.7	67.4	40.2	54.8	62.1	82.6	52.7	43.5	32.9	47.6
2%		WB	21.6	20.9	29.6	34.9	36.1	45.8	54.7	55.5	45.0	35.4	26.5	25.8	
		MCDB	22.5	21.6	32.1	37.9	37.5	48.5	58.1	58.4	47.7	36.4	27.3	27.7	
5%		WB	13.0	13.6	19.2	27.8	34.6	42.2	52.0	51.8	42.5	33.2	23.5	18.2	
		MCDB	13.6	14.5	20.0	29.0	35.8	44.2	54.7	54.0	44.3	33.6	24.6	19.2	
10%		WB	8.4	7.8	11.2	23.4	33.4	39.5	49.3	48.8	40.3	31.4	20.8	13.8	
		MCDB	9.2	8.8	12.2	24.6	34.4	40.8	51.1	50.6	41.5	32.1	21.5	14.8	
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.6	14.4	14.2	8.5	6.6	9.0	7.2	5.5	7.6	10.2	11.0	
		MCDBR	22.2	18.9	21.3	20.3	9.3	13.2	16.7	14.2	11.3	7.5	13.0	16.7	
		MCWBR	21.1	18.6	20.7	18.2	7.7	10.3	13.0	10.8	8.5	6.6	12.5	16.2	
	5% WB	MCDBR	22.0	19.1	21.0	19.6	9.0	12.4	16.2	13.3	10.7	7.2	12.5	16.4	
		MCWBR	20.9	18.8	20.5	17.9	7.6	9.9	13.0	10.5	8.2	6.5	12.1	15.9	
	Clear-Sky Solar Irradiance	taub	N/A	0.202	0.244	0.301	0.327	0.348	0.357	0.339	0.304	0.221	N/A	N/A	
taud		N/A	2.051	2.180	2.087	2.115	2.234	2.380	2.480	2.495	2.179	N/A	N/A		
Ebn at Noon		0	173	246	260	267	266	259	252	229	161	0	0		
Edh at Noon		0	12	23	35	39	36	30	24	18	10	0	0		
All-Sky Solar Radiation	RadAvg	0	118	636	1361	1886	2059	1639	978	521	151	0	0		
	RadStd	0	10	31	76	146	158	137	95	44	19	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.66	N/A	N/A	N/A	N/A	N/A	-603	-626	N/A	N/A

Nomenclature: See separate page